Aayush Patel

Computer Engineering undergraduate building geospatial ML pipelines, backend systems, and data-intensive web applications. Based in Mumbai.

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01
SYSTEMS & WORK

MarineTrace — Oil Spill Detection & Vessel Attribution ↗

SIH 2026 · 2026

Python · Sentinel-1 SAR · Deep Learning · AIS Data · Docker · Render

Automated maritime surveillance pipeline combining synthetic aperture radar (SAR) imagery with real-time vessel telemetry.

- Built a deep learning segmentation pipeline on Sentinel-1 SAR and electro-optical imagery to isolate marine oil slicks from look-alike ocean phenomena.
- Integrated a physical drift engine with spatio-temporal AIS vessel tracking to backtrack spill trajectories and pinpoint candidate source vessels.
- Containerized and deployed the full-stack inference backend and visualization frontend across distributed cloud services.

Pitch-Perfect-Connect / Let's Pitch

Full-Stack Application · 2026

TypeScript · React · Supabase · PostgreSQL · OAuth

Collaborative founder-investor pitch platform with real-time database synchronization and role-based access control.

- Architected relational schemas, row-level security (RLS) policies, and OAuth authentication flows using Supabase and PostgreSQL.
- Designed low-latency pitch deck review workflows and structured feedback channels.

WaterScope

Collaborative Project · 2026

Environmental Monitoring · Geospatial Analytics · Web Stack

Collaborative engineering system focused on water quality telemetry, mapping, and anomaly reporting.

02
SPECIFICATION

Languages	C, C++, Java, Python, SQL, JavaScript / TypeScript, HTML/CSS
ML & Geospatial	SAR Image Segmentation, Geodata Analysis, RAG Pipelines, Hugging Face
Backend & Infra	Supabase, PostgreSQL, JDBC, Docker, Git/GitHub Actions, Render
Core CS	Data Structures & Algorithms, Object-Oriented Programming, Linear & Circular Queues, Linked Lists

03
EDUCATION

B.E. in Computer Engineering

2025 — 2029

Second Year (SY Comp) · Mumbai, Maharashtra

— Coursework: Data Structures, Object-Oriented Programming in Java, Computer Organization.

Additional Technical Training

2026

— **IIRS ISRO Distance Learning:** AI and Machine Learning for Geodata Analysis (Applicant/Participant, Jul 2026).

— **Automation & Retrieval-Augmented Generation (RAG):** Applied LLM workflow architecture.